

Code Break Cipher Relay

Crack ten ciphers and race the answer to the judge before time runs out.

Win condition: Most points across ten codes; a perfect sweep is exactly 100.

THE SCIENCE YOU ARE USING

A cipher is a rule, not magic. Every message in this relay was hidden by one fixed rule: slide the alphabet a few steps (Caesar), mirror it so A trades with Z (Atbash), write it backwards, or hide it in the first letter of each word. Find the rule, run it in reverse, and the message falls out.

Codes swap symbols, not meaning. A message can also trade its letters for numbers (A is 1, Z is 26), Morse dots and dashes, grid coordinates (row digit, then column digit), or five-bit binary. Your decoder sheet holds every table you need; the skill is picking the right tool fast and splitting the work.

HOW TO BUILD AND RUN IT

- 01 Read the slip together.** On GO, flip your team's code slip. The whole team works the code; nobody sits out.
- 02 Pick the right tool.** Decide which rule or table fits: shifted alphabet, mirror, backwards, first letters, numbers, Morse, grid, or binary.
- 03 Split the work.** Divide the symbols among teammates, decode your shares, and assemble the message.
- 04 Run the answer.** One runner whispers the answer to the judge, or shows it written. Wrong answer: the judge says "not yet" and you keep working; unlimited tries, no penalty.
- 05 Rotate and repeat.** A new runner takes the next code. Four minutes per code, ten codes in all.

Safety: Walking feet on the way to the judge; keep the lane between tables clear. Allergy: None.

MATERIALS

Printed decoder sheets	1 per team
Printed code slips	10 rounds, 1 slip per team
Pencils and scratch paper	per team
Timer	judge table

YOUR PLAN AND DATA

A steady team that solves every code beats a fast team that gives up.

Prediction (which code type will be hardest for us, and why): _____

Our runner order (rotate every code): _____

Points after code 5: _____

Points after code 10: _____

DEFEND YOUR DESIGN

What evidence made you change your strategy mid-race? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
First correct team on a code	10
Second to fifth correct team	8, 6, 4, 2
Any later correct answer before time	1
Unsolved when time is called	0
Total · perfect sweep of ten codes	100